Ambulo - Hiking App

Statement Of Work

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Introduction

Nature hikes are an integral part of the leisure experience for people around the world. However, planning a successful trip is a complex task. It requires information from multiple sources like maps, weather apps, and recommendation platforms. The need to consolidate this scattered information and adapt it to individual preferences makes the process time-consuming and complicated.

Additionally, hikers frequently encounter unexpected challenges during their trips, such as confusing trail splits, changing terrain conditions, or sudden weather shifts. Limited access to real-time updates can diminish the experience and, at times, pose safety risks.

The Ambulo project aims to address these issues by providing an innovative, all-in-one solution for planning and navigating hiking trips. Leveraging advanced technologies and artificial intelligence, Ambulo integrates personalized route planning, real-time navigation, and a vibrant community platform for sharing information and experiences.

The app will utilize algorithms to match routes to the user's profile, current weather conditions, and desired difficulty level. It will also offer real-time updates on terrain and trail conditions, points of interest, and tailored equipment recommendations, ensuring a seamless and enjoyable experience for hikers.

Ambulo's data will be sourced from an internal database, external APIs, and user contributions, creating a dynamic and up-to-date platform. The project is structured into three main components:

1. **Algorithm Development** – An algorithm will personalize route suggestions based on user preferences, location, and community recommendations, while also offering equipment guidance.
2. **Client-Side Application** – An intuitive interface will display maps, real-time alerts, and community-shared reviews and experiences.
3. **Server-Side Infrastructure** – A robust backend will handle data processing, user requests, and ensure smooth integration of all features.

Ambulo's mission is to make hiking more accessible, enjoyable, and safe for everyone, allowing users to explore and appreciate nature in a way that best suits their needs.

Goals

To create an accessible, clear, and efficient experience in trip planning and during the trip for all types of travelers, offering a variety of options based on the user and their desires.

Objectives

1. **Personalized Route Selection:** Routes will be recommended based on the user's preferences and hiking history.

2. **Comprehensive Route Descriptions:** Each route will include detailed descriptions, photos, and a list of recommended equipment.

3. **User Progression:** Users will advance through levels, unlocking new permissions, such as editing and creating routes, based on their contributions to the platform.

4. **Weather Forecast Integration:** A weekly weather forecast will be displayed, enabling users to plan ahead effectively.

5. **Seamless Login via OAuth2:** Users can quickly and securely connect to the platform using their Google accounts.

6. **Real-Time Location Features:** During hikes, the application will display the user's current location and provide context-specific alerts, such as tricky trail splits or extreme weather changes.

7. **Tailored Recommendations:** Suggestions will be dynamically generated based on the user's profile and past activities.

8. **Interactive Maps:** The application will feature user-friendly maps showing the current location and route details.

9. **Flexible Route Adjustments:** Users can easily modify routes based on changing preferences, such as proximity to their current location or filtering by region.

Metrics

Personalized Route Selection

- **Metric**: **Route Match Accuracy**
    - Definition: The percentage of recommended routes that align with the user's preferences (e.g., difficulty, distance, terrain type, past activity, etc).
    - Goal: ≥ 80% of routes recommended should align with the user's past behavior and preferences.
- **Metric**: **User Satisfaction with Recommendations**
    - Definition: Rating given by users to the quality of personalized route suggestions on a 5-point scale.
    - Goal: Average rating ≥ 4.2/5.

Comprehensive Route Descriptions

- **Metric**: **Description Completeness - Officials**
    - Definition: The percentage of routes that include all required elements (description, photos, recommended equipment).
    - Goal: 100% of routes in the Officials should include all the required elements.

Weather Forecast Integration

- **Metric**: **Weather Data Accuracy**
    - Definition: The percentage of weather forecasts provided that are consistent with real-time weather conditions.
    - Goal: ≥ 90% accuracy in weather data.
- **Metric**: **User Satisfaction with Weather Forecast**
    - Definition: Rating given by users for the accuracy and usefulness of weather forecasts.
    - Goal: Average rating ≥ 4/5.

Real-Time Location Features

- **Metric**: **Location Accuracy**
  - Definition: The percentage of locations tracked that are within a defined error margin (e.g., 10 meters).
  - Goal: ≥ 95% accuracy of location tracking.
- **Metric**: **Real-Time Alerts Engagement**
  - Definition: Number of times users engage with real-time alerts (e.g., by tapping or acknowledging them).
  - Goal: ≥ 70% engagement rate with real-time alerts.

Literature Review

1. Introduction

Outdoor recreation has become a popular leisure activity, with millions of people exploring nature trails annually. To enhance their experience, hikers often rely on mobile applications for navigation, trail recommendations, and safety information. However, the existing landscape of hiking apps presents gaps in personalization, real-time data integration, and community-driven insights. This review examines relevant applications, technologies, and research that inform the development of Ambulo, a hiking app designed to address these limitations.

2. Review of Existing Applications

Several popular hiking applications cater to outdoor enthusiasts, offering trail maps, user reviews, and navigation tools.

AllTrails: One of the most widely used apps, it features a vast database of trails with user-submitted reviews and photos. Its premium version offers offline maps and real-time overlays. However, its reliance on user-generated content often leads to inconsistencies in trail conditions.

Komoot: Focused on planning and route customization, Komoot integrates cycling and hiking trails but lacks dynamic features like real-time weather updates.

While these apps provide valuable services, they often fall short in integrating real-time alerts, adaptive recommendations, and comprehensive trip planning tools—a gap that Ambulo aims to fill.

3. **Navigation and Mapping Technologies**

Navigation in outdoor applications relies heavily on GPS, digital maps, and routing algorithms. Platforms like Google Maps and OpenStreetMap (OSM) provide foundational data and APIs for app developers.

Google Maps API: Offers precise geolocation and comprehensive routing but focuses on urban and road navigation, with limited support for off-road trails.

OSM: A community-driven mapping project, OSM is highly customizable and widely used for topographic and trail maps. However, maintaining data accuracy requires active community participation.

Ambulo plans to leverage OSM for its trail data, combining it with custom overlays for real-time weather and safety alerts.

4. **Community-Driven Content Platforms**

Apps like **Komoot** and **OffRoad** illustrate the power of user-generated content in enhancing outdoor experiences.**Komoot** allows users to upload and share GPS tracks, fostering a collaborative environment.
Challenges with these platforms include managing data quality and preventing the spread of inaccurate or outdated information. Ambulo will address this by implementing a user ranking system, where experienced contributors gain privileges, ensuring reliable and high-quality content.

5. **Real-Time and Contextual Data**

Dynamic data integration is critical for outdoor safety and convenience. APIs like **OpenWeather** and **IMS (Israeli meteorological system)** provide hyper-local weather forecasts, enabling apps to alert users to potential hazards like storms or extreme heat.

Existing apps often lack seamless integration of such data into their navigation systems. Ambulo will integrate real-time weather data and terrain updates, providing notifications for tricky trail splits or environmental changes, improving both safety and user experience.

6. **Conclusion**

The reviewed literature and applications highlight significant advancements in outdoor navigation and planning. However, gaps remain in personalization, real-time data integration, and community engagement. Ambulo aims to bridge these gaps by combining advanced technologies with user-friendly design, offering hikers a comprehensive tool for planning and executing their outdoor adventures.

Competitors

Category	Parameter	Ambulo	Off Road	Amud Anan	Komoot	AllTrails
Services and Features	Personalized route planning	Yes	Yes	No	Yes	Yes
	Integration of topographic maps and trails	Yes	Yes	Yes	Yes	Yes
	Live alerts for complex points	Yes	No	No	No	No
	Customized equipment lists	Yes	No	No	No	No
	Integrated weather forecasts	Yes	Yes	No	Yes (Only paid)	Yes
	External map integration (Google/OpenStreet Map)	Yes	Yes	Yes	Yes	Yes
User Experience	Simplicity and navigation process (rating from 1-3, 3 is the hardest)	1 (editable)	1 (editable)	3 (Not editable)	2 (Not editable GPX file only)	1 (Not editable GPX file only)
	Multi-language support	No (at this point)	Yes	No	Yes	Yes
Community Engagement	Route ratings	Yes (Starts + filters)	Yes (only based on stars)	No	Yes (only based on stars)	Yes (Starts + filters)

Category	Parameter	Ambulo	Off Road	Amud Anan	Komoot	AllTrails
	Adding reviews and photos	Yes	Yes	Yes	Yes	Yes
	Community-contributed routes	Yes	Yes	No (Only POI)	Yes	Yes
	System based routes	Yes	No	No	No	No
	User levels and special permissions	Yes	No	No	No	No
Technological Innovation	Use of personalized algorithms	Yes	Yes	No	Yes	Yes
	Real-time data integration	Yes	Yes	No	Yes (only paid)	Yes
	Variety of map types	Yes	Yes	Yes	Yes	Yes
Business Model	Free vs. paid versions	Free (at this point)	Free (some maps are paid)	Free (All offline maps are paid)	Free (Most features are paid only)	Mostly paid
	Integrated Ads	Minimal	Minimal	No	Minimal	Highly
	Partnerships (attractions, restaurants)	Not at this point	No	No	Yes	Yes
Support and Maintenance	Regular updates and improvements	Yes	Yes	Minimal (look the same since 2014)	Yes	Yes

detailed requirements

Identify Stakeholders

- **End-Users (Hikers):**
  Individuals who use the Ambulo app to plan and navigate their hiking trips.

- **Project Managers:**
  Oversee project development, ensuring milestones and objectives are met.

- **Developers:**
  Responsible for building, maintaining, and enhancing the Ambulo application.

- **Designers:**
  Create the user interface and ensure a seamless user experience.

- **System Administrators:**
  Manage backend infrastructure, databases, and ensure system reliability.

- **External Partners:**
  Providers of third-party services such as weather data APIs and mapping services.

- **Investors and Stakeholders:**
  Individuals or organizations funding the project and expecting returns or specific outcomes.

- **Regulatory Bodies:**
  Ensure the application complies with relevant laws and regulations, such as data protection standards.

Functional Requirements

1. **User Authentication and Profiles**

Profile Management:

Store user preferences, hiking history, and contribution records.
Allow users to update their profile information, including personal details and hiking preferences.

OAuth2 Integration:

Allow users to register and log in securely using their Google accounts via OAuth2.

2. **Personalized Route Planning**

Route Recommendations:

Utilize algorithms to suggest hiking routes based on user preferences, location, hiking history, and desired difficulty level.

Search and Filter:

Enable users to search for routes by

- Distance
- Region
- Loop
- Includes water sections
- Number of nights
- Trail type, Difficulty level
- Starting point
- Payment required
- Recommended season
- Surface type
- Ending point
- Estimated time
- Wheelchair Friendly
- Dog Friendly

Save and Modify Routes:

Allow users to save favorite routes to their profiles.
Enable users to modify saved routes based on changing preferences or conditions.

3. **Comprehensive Route Descriptions**

Detailed Information:

Provide comprehensive details for each route, including trail length, difficulty level, elevation gain, and estimated hiking time.

Multimedia Content:

Include photos and maps to give users a visual understanding of the route.

Points of Interest:

Highlight significant landmarks, scenic viewpoints, and rest areas along the route.

Equipment Recommendations:

Suggest necessary equipment based on route difficulty and weather conditions.

4. **Interactive Maps**

Real-Time Location Display:

Show the user's current location on the map during hikes.

Route Visualization:

Display the entire hiking route with markers for key points of interest and trail splits.

Map Layers:

Provide options to toggle different map layers such as terrain, satellite view, and trail conditions.

Zoom and Pan:

Allow users to zoom in and out and navigate around the map seamlessly.

5. **Real-Time Alerts**

Trail Condition Updates:

Provide real-time information on trail conditions, including closures, hazards, and maintenance activities.

Weather Alerts:

Notify users of sudden weather changes or extreme conditions during hikes.

6. **Weather Forecast Integration**

Weekly Forecasts:

Display detailed weekly weather forecasts to help users plan their hikes effectively.

Real-Time Weather Updates:

Integrate with external weather APIs to provide up-to-date weather information.

7. **Community Features**

User Reviews and Ratings:

Allow users to leave reviews and rate hiking routes based on their experiences in the community section.

Photo Sharing:

Enable users to upload and share photos from their hikes.

Trail Suggestions:

Let users suggest new trails and provide detailed information for inclusion in the platform.

8. **Equipment Recommendations**

Tailored Suggestions:

Recommend equipment based on the selected route’s difficulty

Customizable Lists:

Allow users to create and customize equipment lists for specific hikes.There will be an option to create a group with other users where the entire trip, including planning, equipment, and more, will be shared among them.

9. **Flexible Route Adjustments**

Dynamic Modifications:

Enable users to adjust their hiking routes in real-time based on changing preferences or unexpected conditions.

Proximity-Based Filtering:

Allow users to filter and adjust routes based on their current location.

Regional Filtering:

Provide options to filter routes by specific districts or areas of interest.

10. **Real-Time Location Features**

GPS Integration:

Utilize device GPS to track and display the user's real-time location on the map.

Context-Specific Alerts:

Provide alerts related to the user's current location, such as upcoming trail splits or areas prone to sudden weather changes.

Navigation Assistance:

Offer turn-by-turn navigation and route guidance to help users stay on track.

11. **Data Synchronization and Offline Access**

Sync Across Devices:

Ensure user data is synchronized across multiple devices for a consistent experience.

Offline Mode:

Allow users to access maps and route information offline, with data synchronization once back online.

Data Caching:

Implement caching mechanisms to store essential data locally for improved performance and reliability.

Non-Functional Requirements

1. **Performance**

Throughput:

The system should handle a minimum of 1000 concurrent users without significant performance degradation.

Resource Utilization:

Optimize server and database resources to handle high traffic efficiently.

2. **Scalability**

Horizontal Scaling:

Design the system to scale horizontally by adding more instances of each microservice to accommodate increasing user loads.

3. **Reliability and Availability**

Uptime:

Achieve an availability target of 99.9%, allowing for scheduled maintenance windows.

Redundancy:

Implement redundant servers and failover mechanisms to minimize downtime in case of component failures.

Disaster Recovery:

Establish disaster recovery plans, including regular data backups and failover strategies.

4. **Security**

Authentication and Authorization:

Implement secure OAuth2 authentication and role-based access control (RBAC) to manage user permissions.

Vulnerability Management:

Conduct regular vulnerability scanning and penetration testing using tools like

5. **Usability**

User Interface:

Design an intuitive and user-friendly interface with clear navigation and accessible design.

Accessibility:

Adhere to WCAG standards to ensure the application is accessible to users with disabilities.

Responsive Design:

Ensure the application is fully responsive and functions seamlessly across various devices and screen sizes.

6. **Maintainability**

Code Quality:

Write clean, modular, and well-documented code to facilitate easy maintenance and updates.

Documentation:

Maintain comprehensive documentation for code, APIs, and system architecture using tools like JSDoc and Swagger.

Version Control:

Use Git with GitHub for effective source code management and collaboration.

7. **Compatibility**

Browser Support:

Ensure compatibility with major browsers, including Chrome, Firefox, Safari, and Edge.

Platform Support:

Support both web and mobile platforms with consistent functionality and performance.

Device Compatibility:

Ensure the application works seamlessly across various devices, including desktops, tablets, and smartphones.

8. **Reliability**

Error Handling:

Implement robust error handling to gracefully manage and recover from system failures.

Automated Alerts:

Set up automated alerts for critical system events and performance issues to enable timely responses.

9. **Performance Optimization**

Database Optimization:

Optimize database queries and use indexing to enhance data retrieval performance.

Load Balancing:

Implement load balancing to distribute traffic evenly across servers, ensuring efficient resource utilization.

10. **Data Management**

Data Integrity:

Maintain data integrity through validation and consistent data handling practices.

Data Backup:

Perform regular data backups and implement retention policies to prevent data loss.

11. **Testing and Quality Assurance**

Automated Testing:

Implement automated testing to ensure code reliability and functionality.

System Architecture

Client-Server Architecture:

Client Side: Web and mobile applications providing user interfaces for hikers.
Server Side: Backend servers handling business logic, data processing, and API requests.
Databases: NoSQL (MongoDB) for structured data and for user-generated content.
APIs: RESTful APIs facilitating communication between client and server.

Technological Requirements

Define key technical requirements:

Browsers: Chrome, Firefox, Edge.
Devices: Desktop, tablet, mobile (Android).
Operating Systems: Windows, Linux, Android.
Frameworks: Flutter for both frontend and backend.

Infrastructure Requirements:

Hosting Platforms: AWS or Azure for backend services.
Servers: Utilize scalable cloud servers with auto-scaling capabilities.
Databases: PostgreSQL for relational data, MongoDB for NoSQL data.
Caching: shared preferences from pub.dev for in-memory caching to enhance performance.

Database Technology:

NoSQL DBMS: MongoDB for flexible data storage.
Scalability: Implement sharding and replication for database scalability.

Cloud Services:

Deployment: AWS Elastic Beanstalk or Azure App Services.
Storage: AWS S3 for user-uploaded content.
CI/CD: Jenkins for automated testing and deployment.

Development Tools:

Version Control: Git with GitHub.
IDE: Visual Studio Code.

Use Cases

Use Case 1: User Registration and Profile Management

Actor: New User

Goal: Register for an account and set up a personalized profile.

Steps:

Registration:

User navigates to the Ambulo app's registration page.
User selects "Sign up with Google" to initiate OAuth2 authentication.
Users grant necessary permissions for Ambulo to access their Google account information.
System creates a new user account and redirects the user to the profile setup page.

Profile Setup:

User input additional information such as hiking preferences, favorite trail types, and personal details.
User saves the profile information.

Possible Results:

Success: User successfully registers and sets up their profile, gaining access to personalized features.
Failure: If OAuth2 authentication fails, the system notifies the user and prompts retrying the registration process.

Use Case 2: Plan a Personalized Hiking Trip

Actor: Registered User

Goal: Create and save a personalized hiking trip based on preferences and current conditions.

Steps:

Input Preferences:

User access the "Plan a Trip" feature in the Ambulo app.
User selects preferences such as desired location, difficulty level, trail length, and preferred hiking season.

Receive Recommendations:

System processes the input using recommendation algorithms.
System displays a list of suggested hiking routes that match the user's criteria and current weather conditions and other information.

Review and Select Route:

Users review detailed descriptions, photos, and equipment recommendations for each suggested route.
User selects a preferred route and there will be two option - start, save

Save and Modify:

System saves the selected route to the user's favorite.
Users can later modify the saved route based on changing preferences or new conditions.

Possible Results:

Success: User successfully plans and saves a personalized hiking trip.
Failure: No routes match the user's criteria, prompting the user to adjust their preferences and suggesting alternative routes

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Use Case 3: Navigate a Hike with Real-Time Alerts

Actor: Active Hiker

Goal: Navigate a hiking trail with real-time updates and alerts for safety and convenience.

Steps:

Start Navigation:

User selects a saved hiking route and clicks "Start Hike."
System activates GPS tracking and displays the current location on the interactive map.

Receive Real-Time Alerts:

As the hike progresses, the system sends real-time alerts about trail conditions, weather changes, and upcoming tricky trail splits.
Users receive push notifications and in-app alerts based on their location and predefined preferences.

Emergency Assistance:

In case of an emergency, user can access quick links to emergency contacts and safety instructions directly from the alerts.

Possible Results:

Success: User navigates the trail safely with timely alerts and assistance.
Failure: If GPS signal is lost, the system notifies the user and attempts to reconnect.

Use Case 4: Share Hiking Experience in Community

Actor: Registered User

Goal: Share reviews, photos, and suggestions to contribute to the Ambulo community.

Steps:

Create a Review:

After completing a hike, user accesses the "Community" section.
User select the completed route and write a review detailing their experience, including ratings for difficulty, scenery, and trail conditions.

Upload Photos:

User uploads photos taken during the hike to accompany the review.
Photos are tagged with location and time for better context.

Suggest a New Trail:

User selects the option to suggest a new trail if they explored an unmapped area.
User provides detailed information about the new trail, including trailhead location, length, difficulty, and points of interest.

Engage with Community:

Other users can view, like, and comment on the reviews and photos.
User receives.

Possible Results:

Success: Users successfully share their hiking experience, contributing valuable information to the community.
Failure: If the suggested trail lacks necessary details, the system prompts the user to provide additional information.
